

WINE QUALITY PREDICTION - EDA REPORT

DATASET OVERVIEW

Original Records: 1599

Original Columns: 12

Records After Duplicate Removal: 1359

Input Features: 11

Target Variable: quality

Missing Values: 0

Duplicate Records Found: 240

Numerical Features: 12

Categorical Features: 0

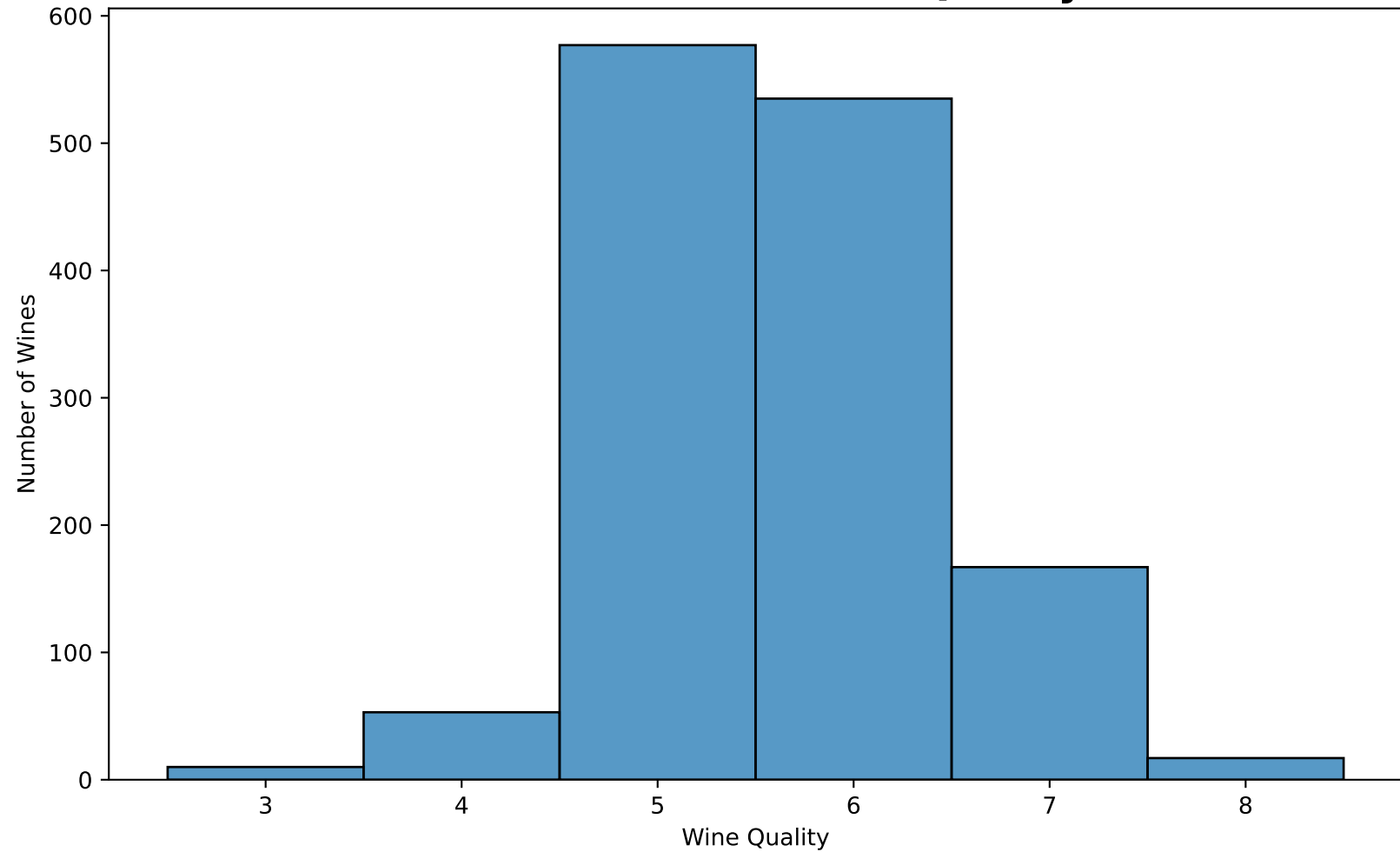
Quality Values:

[np.int64(3), np.int64(4), np.int64(5), np.int64(6), np.int64(7), np.int64(8)]

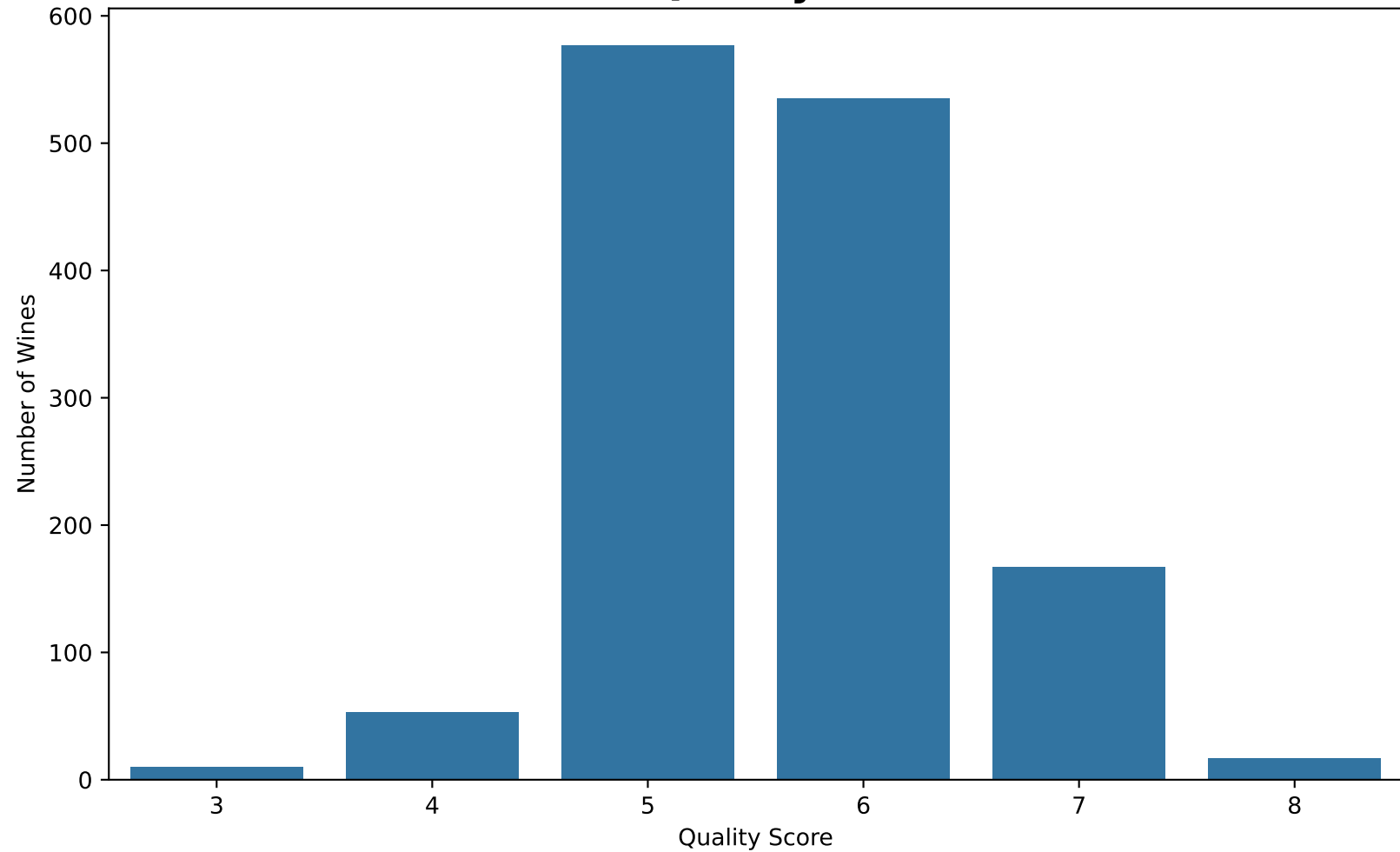
Statistical Summary

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
count	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0	1359.0
mean	8.311	0.529	0.272	2.523	0.088	15.893	46.826	0.997	3.31	0.659	10.432	5.623
std	1.737	0.183	0.196	1.352	0.049	10.447	33.409	0.002	0.155	0.171	1.082	0.824
min	4.6	0.12	0.0	0.9	0.012	1.0	6.0	0.99	2.74	0.33	8.4	3.0
25%	7.1	0.39	0.09	1.9	0.07	7.0	22.0	0.996	3.21	0.55	9.5	5.0
50%	7.9	0.52	0.26	2.2	0.079	14.0	38.0	0.997	3.31	0.62	10.2	6.0
75%	9.2	0.64	0.43	2.6	0.091	21.0	63.0	0.998	3.4	0.73	11.1	6.0
max	15.9	1.58	1.0	15.5	0.611	72.0	289.0	1.004	4.01	2.0	14.9	8.0

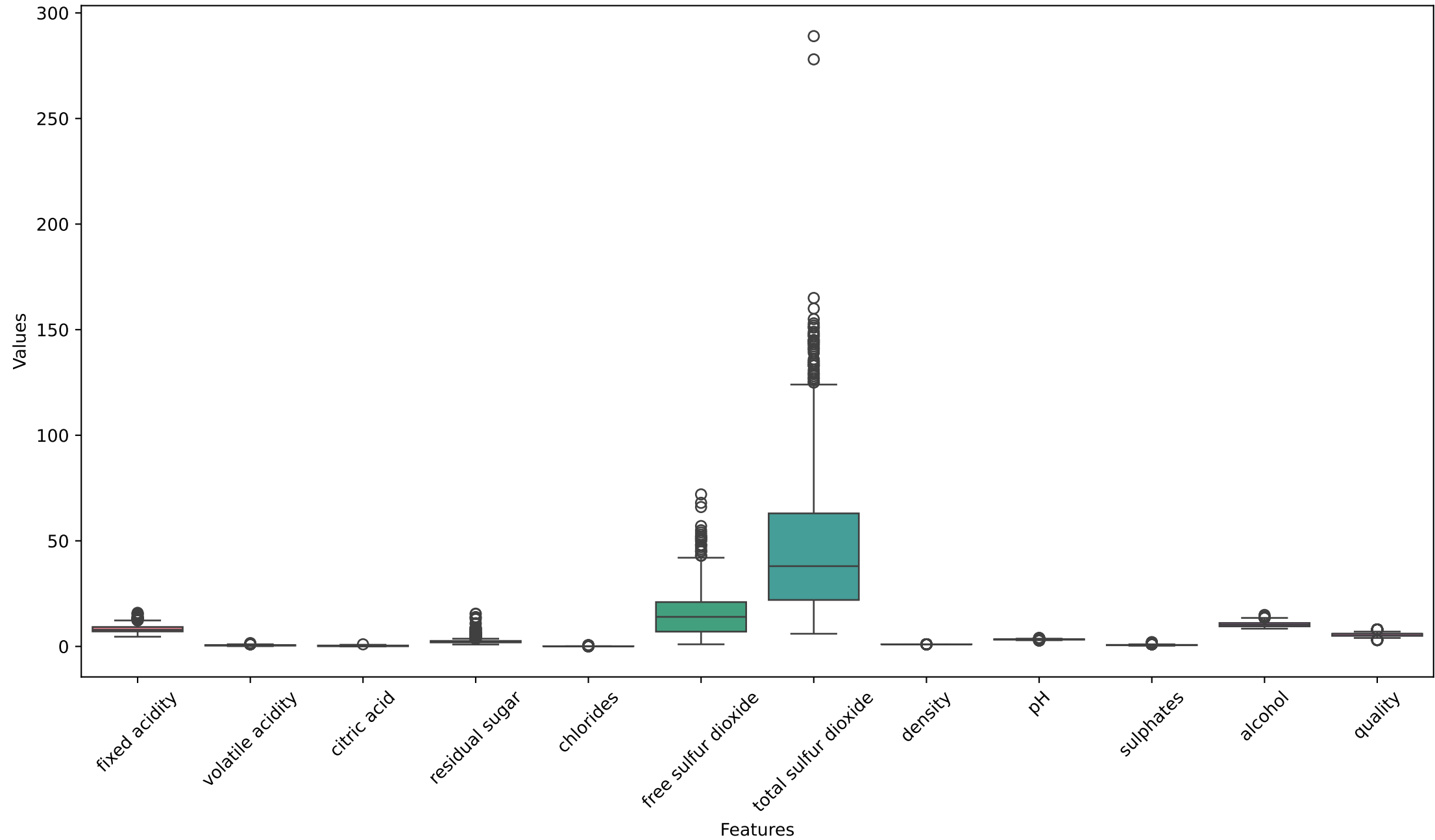
Distribution of Wine Quality



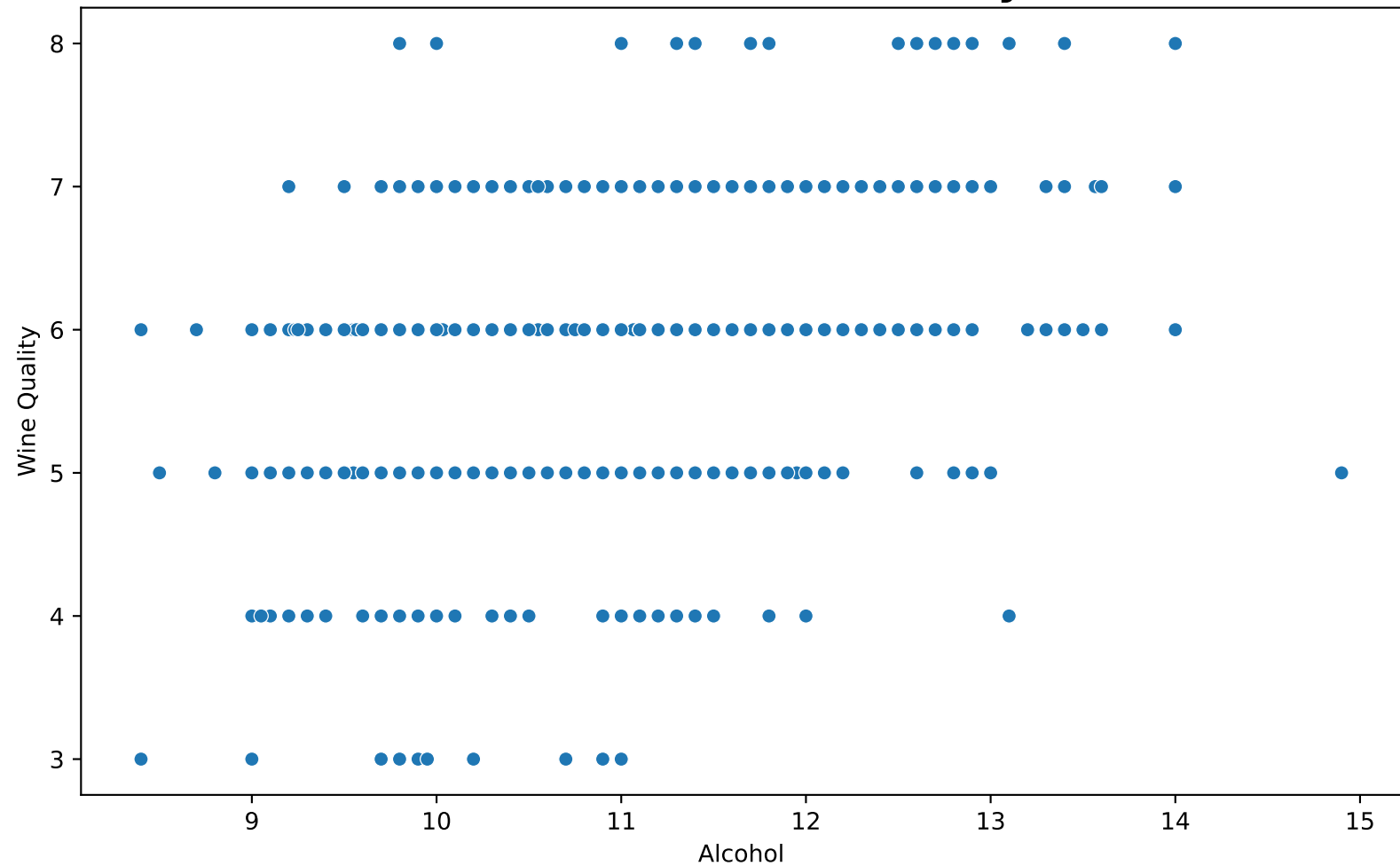
Wine Quality Counts



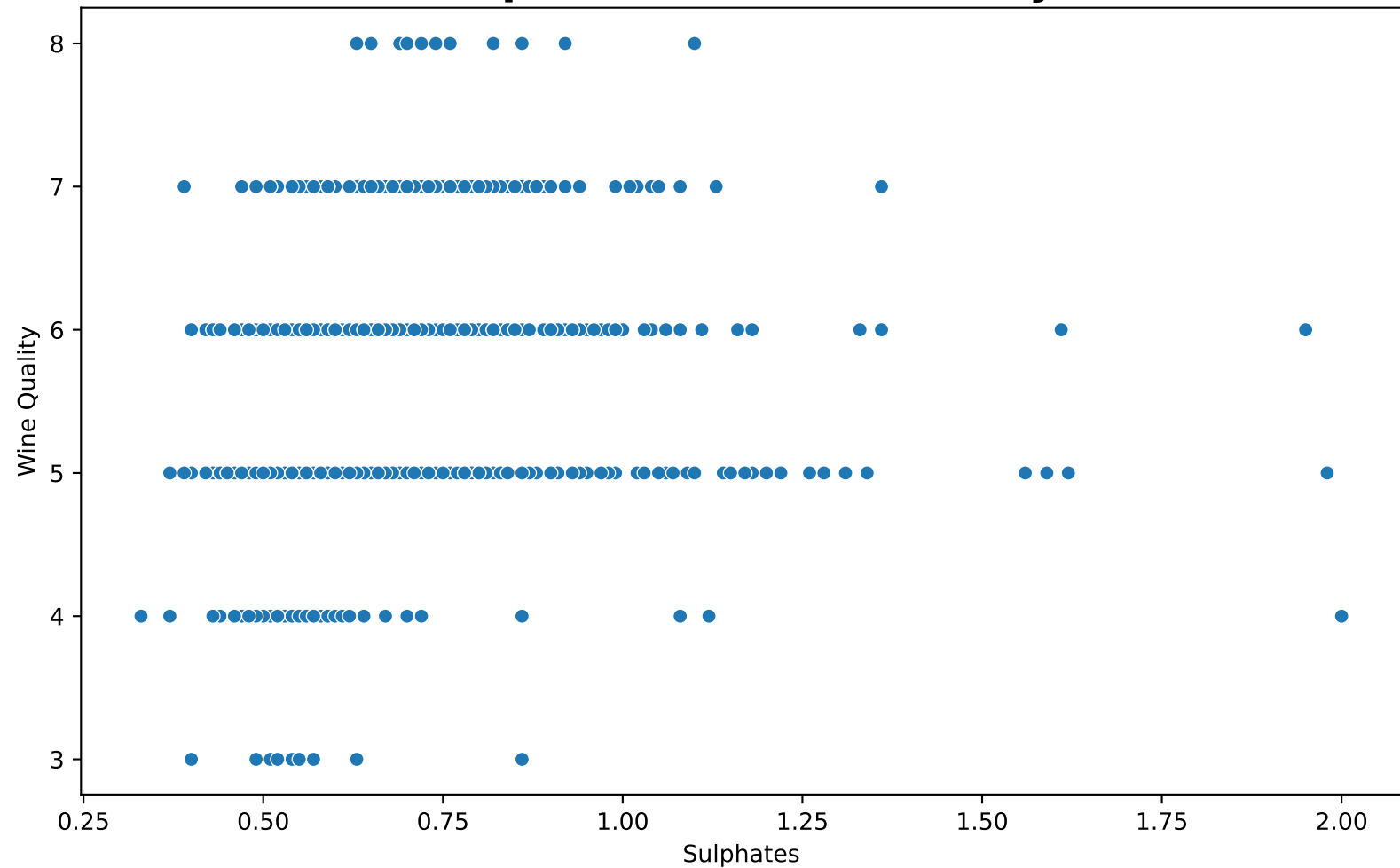
Box Plot of Wine Chemical Features



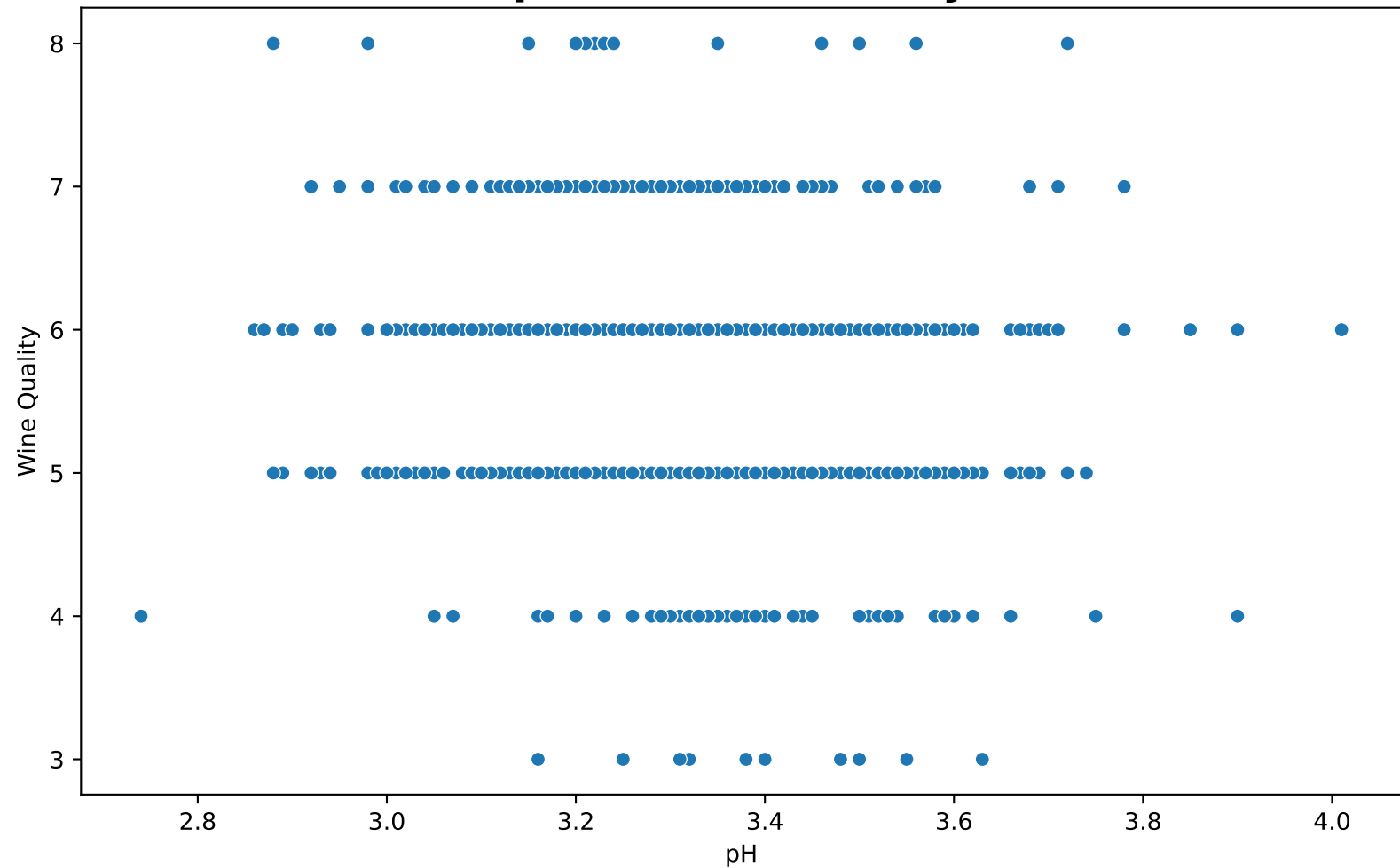
Alcohol vs Wine Quality



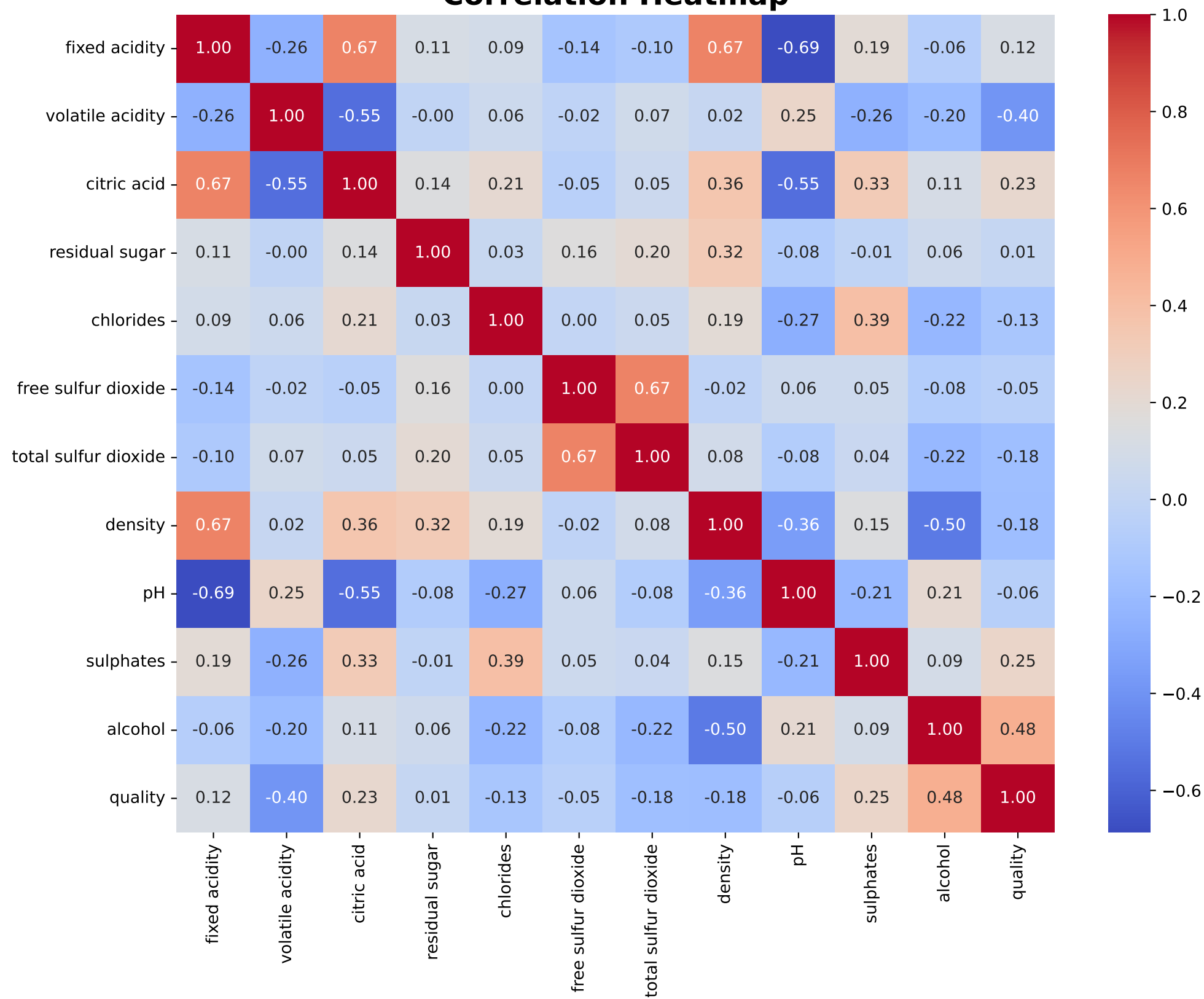
Sulphates vs Wine Quality



pH vs Wine Quality



Correlation Heatmap



EDA Findings and Conclusion

EDA OBSERVATIONS AND CONCLUSION

- The cleaned dataset contains 1359 wine records and 11 input chemical features.
- No missing values were found.
- 240 duplicate records were identified and removed during preprocessing.
- Wine quality values range from 3 to 8.
- Quality scores 5 and 6 occur most frequently.
- Box plots were used to identify potential outliers.
- Scatter plots were used to study relationships between chemical properties and wine quality.
- The correlation heatmap shows relationships between the chemical features and the target variable.
- All input features are numerical.

CONCLUSION

EDA helped us understand the distribution, structure, relationships, and possible outliers in the Wine Quality dataset. These findings were used to prepare the data for Ridge, Lasso, and Elastic Net regression models.